Question Statement

Which of the following best describes a process in an operating system?

What is the main objective of scheduling algorithms in OS?

Which scheduling policy selects the process that arrives first?

What does SJF stand for in process scheduling?

In STCF, a process with ____ is scheduled next.

Which scheduling algorithm assigns time slices to each process?

Which of these is not a process scheduling metric?

Turnaround time is calculated as:

Which scheduling method favors shorter jobs?

In Priority Scheduling, the CPU is assigned to the process with:

Round Robin scheduling is most suitable for:

Which one is not a necessary condition for deadlock?

Which of the following can lead to a deadlock?

Deadlock prevention can be achieved by eliminating:

Deadlock avoidance is implemented using which method?

Which algorithm is used for deadlock avoidance?

What happens in a circular wait deadlock?

What is a safe state in deadlock avoidance?

Which of the following is a non-preemptive scheduling algorithm?

Which scheduling metric measures the time a process spends waiting in the ready queue?

Which scheduling policy allows a running process to be interrupted?

In Priority Scheduling, which process gets scheduled first if two processes have the same priority?

In MLFQ, a process using more CPU is moved to:

Which of the following is an example of starvation?

What defines Response Time in a process?

A job arrives at time 2 ms and completes at 12 ms. What is the turnaround time?

Process P1 needs 4 ms CPU time, P2 needs 3 ms CPU time, both arrive at same time. FCFS average waiting time?

In SJF, processes have burst times 5ms, 3ms, 8ms. Average turnaround time is closest to:

Given 3 processes with arrival times (0ms, 2ms, 4ms) and burst times (5ms each), average waiting time with FCFS?

In RR with quantum 4 ms, P1 burst=10ms. How many times is P1 scheduled before completion?

In Priority Scheduling, if P1(priority=2) and P2(priority=1), which executes first?

Deadlock can occur if system has ___ resources and ___ processes.

Process P1 arrives at 0ms needs 7ms CPU, P2 arrives at 2ms needs 4ms CPU. In FCFS, what is average turnaround time?

If Turnaround Time = 20ms, Waiting Time = 12ms, what is CPU burst time?
What is the main function of the main memory?
Which memory allocation technique suffers from external fragmentation?
Which dynamic memory allocation algorithm searches the entire list for the smallest free suitable partition?
In contiguous memory allocation, which method allocates memory blocks first that fit the process size?
Paging helps in solving which of the following problems?
The mapping from logical to physical address is done by:
Which algorithm replaces the page that has not been used for the longest period of time?
In FIFO page replacement, which page is removed?
A system has 4 frames. A page reference string 1,2,3,4,1,2,5. How many page faults with FIFO?
If a system uses LRU replacement, the least recently accessed page is:
Which of the following is a fixed partition allocation scheme?
The main advantage of dynamic partitioning over fixed partitioning is:
External fragmentation occurs in:
In FIFO page replacement, increasing number of frames sometimes increases page faults. This is called:
Which memory management scheme requires relocation of processes during execution?
Page replacement occurs when:
In contiguous memory allocation, which strategy always allocates the largest available block?
In paging, what is the main advantage compared to contiguous memory allocation?
The basic unit of data exchange between secondary memory and main memory is called:
What is concurrency in operating systems?
Which of the following helps manage concurrency?
A lock is used to:
What is a critical section?
Which of the following is NOT a lock-based synchronization mechanism?
What is a mutex?
When does a thread spin in a spinlock?
Which condition must be satisfied for mutual exclusion?
Which function is commonly associated with condition variables?
Which two operations are associated with condition variables?
What is a semaphore?
A binary semaphore can take values:
What is a file system interface responsible for?
Which operation creates a new file?

Which system call is used to open a file in C?
Which system call reads data from a file?
Which system call writes data to a file?
The fork() system call is used to:
The dup() system call is used to:
Which command deletes a directory?
Which system call removes a file in UNIX?
What happens to open file descriptors after fork()?
Which operation lists files in a directory?
Which data structure stores metadata about a file in UNIX?
Which operation must happen before writing to a file?
Which of the following is used to organize files?
Which is NOT a disk scheduling algorithm?
Which algorithm services request closest to current head position?
Purpose of disk scheduling is:
Which scheduling moves head from one end to another and back?
What does SCAN algorithm do?
Why is C-SCAN better than SCAN?
Which algorithm gives best average seek time?

Option One	Option Two
Program in execution	Stored program
Maximize CPU utilization	Minimize memory usage
Shortest Job First	First-Come First-Served
Simple Job Finish	Shortest Job First
Longest remaining time	Shortest remaining time
Priority Scheduling	Round Robin
Turnaround Time	Waiting Time
Finish Time - Arrival Time	Arrival Time - Start Time
FIFO	SJF
Lowest ID	Highest priority
Batch systems	Real-time systems
Mutual Exclusion	Hold and Wait
Mutual exclusion	Preemption
Hold and Wait condition	All threads
Resource Allocation Graph	Bankers Algorithm
Banker's Algorithm	Dijkstra's Algorithm
Each process waits for a resource held by next process	A resource is not released
A state where deadlock is impossible	A state where deadlock has occurred
Round Robin	Priority Scheduling
Turnaround Time	Waiting Time
Non-preemptive	Preemptive
First entered	Last entered
Lower priority queue	Higher priority queue
A low-priority process waiting indefinitely	A deadlock situation
First Response - Arrival Time	First Response - Completion Time
10 ms	12 ms
1.5 ms	2 ms
9 ms	10 ms
5 ms	4 ms

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P1	P2
More, Less	Equal, Equal
10ms	13ms

8ms	10ms
Store data permanently	Temporary storage of data and instructions
Paging	Segmentation
First Fit	Best Fit
First Fit	Best Fit
External Fragmentation	Internal Fragmentation
CPU	Operating System
FIFO	LRU
Oldest loaded page	Least used page

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Replaced	Locked
Paging	Segmentation
Less complexity	Less fragmentation
Paging	Contiguous Memory Allocation
Thrashing	Belady's Anomaly
Segmentation	Swapping
A process requests a page not in memory	Page is removed from main memory
First Fit	Best Fit
Eliminates internal fragmentation	Increases execution speed
Segment	Frame
Multiple tasks running simultaneously	Single task execution
Deadlock	Lock
Prevent unauthorized access	Ensure exclusive access to a shared resource
A section of code accessing shared resources	A program header
Spinlock	Mutex
A mutual exclusion object	A memory address
While waiting to acquire the lock	When releasing the lock
Only one thread can access the critical section at a time	Multiple threads access simultaneously
wait()	execute()
lock and unlock	wait and signal
An encryption algorithm	A scheduling method
Any integer	0 and 1
Managing files and directories	Memory management
mkdir	create

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open()	close()
read()	open()
write()	read()
Create new processes	Duplicate a file descriptor
Copy file content	Create a process
rmdir	mkdir
unlink()	open()
They are duplicated	They are copied independently
rmdir	mkdir
data block	inode
Opening the file	Deleting the file
Superblock	inode
SSTF	Round Robin
SSTF	FCFS
Increase memory	Maximize CPU load
LOOK	SSTF
Moves disk head in one direction then back	Jumps to first request
Provides uniform wait time	Moves faster
SSTF	SCAN

Option Three	Option Four	Correct Option
Instruction	Procedure	A
Increase system calls	Reduce network latency	A
Priority Scheduling	Round Robin	B
Shortest Jump First	Simple Job First	B
Highest priority	Lowest priority	B
MLFQ	First-Come First-Served	B
Response Time	Execution Instruction	D
Start Time - Finish Time	Finish Time - Execution Time	A
Priority Scheduling	Round Robin	B
Lowest priority	Longest execution time	B
Time-sharing systems	Multiprocessing systems	C
No Preemption	Interruptible execution	D
Spooling	Thread yielding	A
Semaphores	Memory paging	A
Spinlocks	Mutex	B
Kruskal's Algorithm	Floyd's Algorithm	A
A process locks two resources	Resources are infinitely produced	A
A state where processes are terminated	A state with all resources free	A
First-Come First-Served	MLFQ	C
Response Time	Execution Time	B
Sequential	Serial	B
Shortest job	Longest job	A
Suspended state	Blocked state	A
A preemptive switch	A blocked thread	A
First CPU Allocation - Arrival Time	Execution Time	A
8 ms	6 ms	A
3 ms	3.5 ms	B
11 ms	12 ms	B
6 ms	7 ms	B
4	5	B
Random	Depends on Arrival	B
Less, More	None	C
12ms	11.5ms	D

12ms		14ms	A
Perform calculations		Manage I/O devices	B
Contiguous Memory Allocation		Virtual Memory	C
Worst Fit		Next Fit	B
Worst Fit		Next Fit	A
Both		None	A
Memory Management Unit		Compiler	C
Optimal		NONE	B
Random page		Smallest page	A
	6		7 B
Moved		Pinned	A
Contiguous Allocation		Virtual Memory	A
Better utilization		Less memory requirement	C
Virtual Memory		Pure Demand Paging	B
Overhead		Fragmentation	B
Paging		Contiguous Allocation	B
A page is loaded into memory from disk		Frame is locked	A
Worst Fit		Next Fit	C
Reduces page faults		Eliminates external fragmentation	D
Block		Page	D
Single-threaded processing		Sequential processing	A
Cache		Virtual memory	B
Increase CPU speed		Manage disk scheduling	B
A memory location		A CPU register	A
Semaphore		Cache	D
An interrupt handler		A paging algorithm	A
When scheduling		While executing critical section	A
All threads must sleep		Threads must terminate	A
release()		fork()	A
create and destroy		read and write	B
A type of memory		A signaling mechanism	D
0 and 2		1 and 2	B
Network communication		Managing disk blocks	B
link		open	B

fork()	unlink()	A
write()	fork()	A
fork()	dup()	A
Clone a memory segment	Copy file content	A
Create a new memory block		D
unlink	cd	A
mkdir()	link()	A
They are closed	They are deleted	A
rm	ls	D
file content	buffer	B
Forking a process	Closing another file	A
Directory	Scheduler	C
LOOK	SCAN	B
SCAN	LOOK	A
Reduce seek time	Save bandwidth	C
SCAN	FCFS	C
Moves head randomly	Only moves clockwise	A
Reduces CPU load	Skips requests	A
C-SCAN	LOOK	A